

Intone your Talk: Voice Output Communication Aids with Individualized Voices for Augmented Communicators

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Abstract

Voice Output Communication Aids (VOCAs) enable communication for people who cannot use their natural voices, yet they offer limited opportunities for voice personalization. This poster presents an exploratory qualitative study of how augmented communicators experience voice personalization and engage in shaping synthetic voices. We conducted semi-structured interviews with six VOCA users, complemented by hands-on interaction with VoiceX, an AI-based design probe for iterative voice exploration. Participants expressed a desire for self-representative voices but often struggled to articulate such preferences. Interaction with VoiceX revealed a tension between exploratory freedom and perceived agency. At the same time, the system supported diverse communication habits through flexible interaction design, pointing to directions for future AAC research.

CCS Concepts

- **Human-centered computing** → HCI design and evaluation methods; Participatory design; Interface design prototyping;
- **Applied computing** → Industry and manufacturing.

Keywords

Human-Centered AI, co-creation, Human-Robot interaction

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1 Introduction

Imagine a classroom in which several students who cannot use their natural voices rely on Voice Output Communication Aids (VOCAs) to participate in a lecture. Although these devices enable communication, they often produce similar-sounding synthetic speech, making it difficult to convey individuality and distinguish speakers. While orally communicative individuals express identity through their voices, augmented communicators, i.e., users of Alternative and Augmentative Communication (AAC), have limited opportunities to do so through the vocal channel.

Despite ongoing advances in AAC technology, most VOCAs offer only a small set of predefined synthetic voices that rarely reflect users' age, gender, emotions, or personality. This lack of personalization can hinder speaker differentiation and self-expression [13] and contribute to a reduced sense of vocal persona [8]. Rather than treating voice as a neutral output modality, this work examines how voice personalization is experienced by augmented communicators in practice.

We address this through a qualitative interview study with AAC users, complemented by hands-on interaction with VoiceX, a tool used here as a design probe for iterative voice exploration [14]. Interviews were flexibly adapted to participants' communication practices and supported by familiar mediators where necessary [3, 12]. By combining reflective interviews with interactive voice exploration, we aim to understand how users relate personalized synthetic voices to identity, agency, and control—without claiming generalizable design solutions.

2 Related Work

Research on AAC has long emphasized that voice is not merely a channel for transmitting content, but a key resource for identity, self-expression, and social positioning. Kane et al. [6] show that limited control over synthesized speech restricts users' ability to convey personality and intent, resulting in reduced opportunities for self-expression. Similarly, Pullin's design research [9, 10] foregrounds tone of voice as a social and emotional medium, using

speculative and interactive artefacts to provoke reflection on how voices shape interpersonal interaction in AAC.

Across this body of work, a recurring tension emerges between expressivity and control. While greater control over vocal parameters can enable nuanced expression, it can also introduce cognitive and interactional burdens that disproportionately affect augmented communicators. Prior systems therefore often balance expressivity through constrained interfaces, offering predefined voices or limited modulation rather than open-ended personalization [2, 5]. This trade-off has shaped much of the design space for AAC voices.

Recent advances in speech technology have renewed interest in long-term voice personalization. Voice banking and voice cloning approaches aim to preserve or reconstruct an individual's voice, particularly for people with degenerative conditions [1, 4, 7]. While such approaches can support continuity of identity, they typically prioritize replication of an existing voice and assume access to suitable speech data. As a result, they may be less applicable to users who have never spoken or who wish to explore alternative vocal identities [13].

Our work builds on this distinction by shifting the focus from replication to exploration. Rather than treating personalization as the recovery of a "true" voice, we investigate how augmented communicators experience the process of shaping a synthetic voice through interaction. Using VoiceX as a design probe [14], we complement prior work on expressive AAC voices by examining how agency, identity, and control are negotiated during iterative voice exploration, particularly when personalization is mediated by AI-driven generation rather than direct parameter manipulation.

3 The VoiceX System

The aim of our work is to understand how augmented communicators approach voice personalization. Following recommendations for qualitative research with AAC users [3, 12], we employed a tangible elicitation tool that allowed participants to actively explore synthesized voices rather than only describe preferences verbally. To this end, we used VoiceX [14], a system that enables users to explore and shape synthetic voices through lightweight interaction. VoiceX is used here exclusively as a design probe to prompt reflection, not as a validated or deployable AAC system. It served to support hands-on exploration of voice personalization and to elicit participants' reactions, strategies, and expectations during the process.

At a high level, VoiceX enables iterative exploration of synthetic voices through a sequence of simple comparisons. Participants listen to two candidate voices and indicate which one they prefer (see Fig. 1). Based on this selection, the system generates a new pair of voices for the next iteration. This interaction enables voice exploration without requiring technical or acoustic specifications. Underlying this interaction is a user-in-the-loop evolutionary search process. VoiceX represents voices as latent speaker embeddings in a continuous space. At each iteration, new candidate voices are sampled by slightly mutating and recombining embeddings derived from previously selected voices. Participant choices thus function as an implicit preference signal, steering the exploration toward regions of the voice space that better match their evolving sense of

what "fits." This approach balances exposure to diverse voice variations with gradual preference-guided narrowing, while keeping the interaction simple and accessible.

Interaction with VoiceX is mediated through a minimal web-based interface (Fig. 1). At each step, participants are presented with two buttons corresponding to the candidate voices and a third option to stop the process once they feel satisfied. This interaction pattern supports diverse communication strategies, including verbal responses, VOCA-mediated commands, gaze-based signaling, or mediation by accompanying persons.

VoiceX was deliberately chosen as a lightweight and flexible probe that foregrounds auditory experience while keeping interaction demands low. This allowed participants to focus on evaluating voice qualities, such as perceived naturalness, tone, or familiarity, while revealing how they reason about preference, control, and authorship during voice creation. Accordingly, our analysis focuses on participants' interactions with the probe and their interpretations of its behavior, rather than on technical performance or optimization quality, to understand how voice personalization is experienced in AAC.

4 Study

We conducted an exploratory qualitative study with six augmented communicators (aged 22–54; 3 female, 3 male). Participants included four individuals with cerebral palsy (P1, P2, P5, P6) using a VOCA in combination with an eye tracker and two with intellectual disabilities (P3, P4) using a VOCA in combination with an IPAD. The study examined how augmented communicators experience current VOCA voices and how they engage with voice personalization when supported by an interactive design probe.

The study is explicitly exploratory. Participants differed in communication practices, cognitive and physical abilities, and familiarity with technology, which shaped how they engaged with the interviews and the VoiceX probe. We do not compare participants across diagnoses; instead, we treat each interaction as a situated account of voice personalization. Accordingly, findings are interpreted case-wise rather than as representative or comparative across groups.

Participants were recruited through organizations supporting people with disabilities. Interviews were adapted to participants' preferences regarding location, timing, and pacing, and were conducted in familiar settings such as homes, workplaces, or a university environment. Written informed consent was obtained from participants or their legal guardians.

Interviews were designed to accommodate diverse communication needs. Participants were encouraged to attend with a trusted communication partner, and most did so. Responses were expressed through multiple modalities, including VOCA-mediated utterances, brief verbal responses, non-verbal cues, and mediated communication via accompanying persons. Interviewers flexibly adapted question structure and pacing to reduce fatigue and support participation, following established AAC research practices [3, 12].

The interview protocol covered participants' experiences with current VOCAs, perceptions of voice output, and expectations regarding voice personalization. In the second part of each session, participants interacted with VoiceX to explore synthetic voice creation (see Fig. 2). VoiceX was introduced as an exploratory design

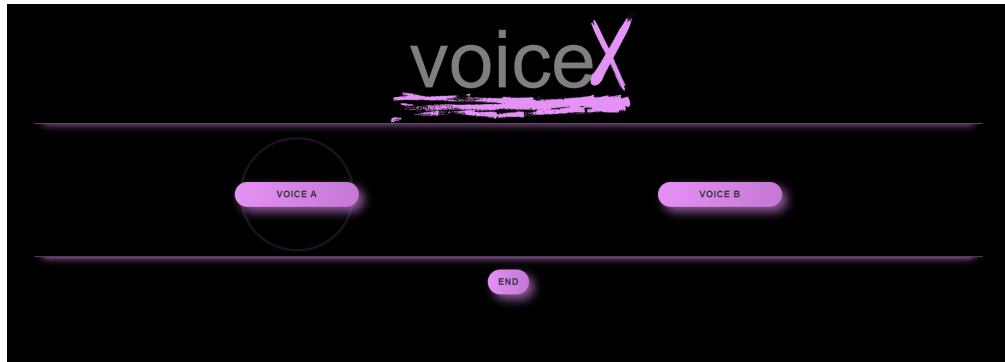


Figure 1: The VoiceX interface. Two buttons are shown to select the better voice of the current iteration. After selecting that voice, the new iteration is synthesized. The third button at the bottom of the interface can be used to end the process as soon as the participant is happy with one voice of the current iteration.

probe rather than a deployable system, and participants were free to pursue any voice characteristics or goals they found meaningful. Interaction with the system was supported by a research team member and, where needed, by the accompanying person. Each session lasted approximately 60 minutes.

Data analysis followed qualitative content analysis [11]. Interviews were audio-recorded, transcribed, and pseudonymized. Spoken, VOCA-mediated, and non-verbal contributions were documented, including whether utterances originated directly from participants or were mediated by others. The coding scheme was iteratively refined to capture emerging themes beyond the initial interview structure. All coding was conducted using MAXQDA 2024.

5 Results

We present four thematic findings that synthesize interview data and participants’ interactions with VoiceX. Rather than cataloguing device features or usage patterns, we focus on how participants experienced voice personalization in relation to identity, agency, and control. VoiceX served as a design probe to elicit reflection and action, complementing verbal accounts with hands-on exploration.

5.1 Theme 1: Voice as Identity - Desired but Hard to Articulate

Participants reported limitations of their current VOCA voices, including reliance on standard voices and limited emotional expressiveness. Against this background, participants consistently framed voice as central to identity and self-expression, yet struggled to articulate what their “own” voice should sound like. P2 summarized this tension succinctly: “Ich möchte mich.” (“I want me.”), while simultaneously acknowledging the challenge of defining such a voice: “Ist schwer zu erklären, habe noch nie selber gesprochen.” (“It’s difficult to explain, I’ve never spoken myself.”). This combination of desire and resignation suggests that limited personalization options shape not only what voices users have, but also what they consider conceivable. Notably, not all participants aimed to create

a self-representative voice. Some were equally interested in experimenting with alternative or familiar voice styles, indicating that personalization does not necessarily equate to identity replication.

5.2 Theme 2: Agency: Self-Directed or Mediated

Participants’ interactions with VoiceX showed that agency emerged through interaction, rather than being built into the system. Several participants expressed a strong desire to independently shape their voices, emphasizing ownership over the process. P5 stated unambiguously: “Natürlich alles selber machen.” (“Of course, I’ll do everything myself.”). Others were less certain. P3 preferred her mother to create her voice, illustrating that agency in AAC contexts can be relational and mediated rather than purely individual. For participants who engaged less directly with the interface, such as P3 and P4, accompanying persons played a substantial role in mediating interaction. Here, caregivers effectively served as co-authors in the voice-creation process, shaping selections and pacing. Importantly, this mediation did not erase participant preferences; subtle but consistent choices (e.g., favoring softer voices) indicated that agency could still be expressed indirectly. Overall, agency emerged as distributed across users, mediators, and system behavior.

5.3 Theme 3: Exploration vs. Control: VoiceX as Productive Tension

Participants adopted different exploration styles, ranging from rapid, intuitive selection to careful approximation toward a preferred voice. For some, the process was engaging and playful. For others, it was goal-directed. Participants expressed varied perspectives on the extent of control they experienced in the process of voice selection and customization with VoiceX. For some, the AI techniques at the core of VoiceX clearly supported a strong sense of agency. P5, for instance, emphasized that he felt fully in control of the system, noting that his choices directly shaped the resulting voice. In some cases, participants experienced a lack of control when generated voices failed to align with their intentions, particularly when the goal was self-representation. P2 rejected the outcome clearly: “Ich. Bin. Nicht. Ich. Mich.” (“I. Am. Not. Myself.”). Rather



Figure 2: Participants using VoiceX on their own (left) or facilitated by their communication partner (right)

than disengaging, such mismatches often motivated further customization, with P2 and P6 explicitly expressing a desire to create their own voices in the future.

5.4 Theme 4: Interface Legibility: Simple Control, but Complex Inner Workings

VoiceX enabled participants to explore synthetic voices through a minimalistic, binary-choice interface that supported diverse communication strategies. P1 communicated which option to choose via gaze commands, which his personal assistant spontaneously suggested before the interaction with VoiceX started. Based on these commands, the assistant controlled VoiceX. The other participants used short verbal utterances, such as "Die da" ("This one"), when selecting voices. Despite its simplicity, the interface did not make the generative process easy to understand or immediately to converge: participants were often unsure how their selections influenced subsequent outputs. Consequently, the number of iterations needed to achieve a satisfactory voice varied widely, ranging from 3 (P6) to more than 20 (P1).

6 Discussion and Limitations

This work highlights that voice personalization in AAC is not experienced as a purely technical task, but as a situated and interactional process shaped by identity, agency, mediation, and system constraints. Across interviews and interactions with VoiceX, participants framed voice as central to self-expression, while simultaneously revealing how difficult it can be to articulate what an "own" voice should sound like - particularly for individuals who have never spoken themselves. Rather than converging on a single, stable goal, voice personalization emerged as an exploratory process involving experimentation, approximation, and sometimes resignation.

Interaction with VoiceX, used here as a design probe, made visible how agency in voice creation is negotiated rather than given. Some participants experienced moments of control and ownership when their selections appeared to meaningfully shape the generated voices, while others encountered dependency on caregivers or

uncertainty about how their choices influenced system behavior. Caregiver mediation often enabled participation, but also shaped pacing and interpretation, underscoring that voice authorship in AAC is frequently distributed across users, mediators, and technology.

The minimal, binary interaction pattern of VoiceX supported participation across diverse communication practices and enabled hands-on exploration of synthetic voices. At the same time, participants' hesitations when voices sounded similar, or when neither option was appealing, revealed a productive tension between exploratory freedom and perceived control. Reduced interaction complexity can facilitate access, but does not necessarily enhance transparency or predictability of the underlying generative process. These experiences should not be understood as system failures, but as insights into how personalization is encountered in practice.

Several limitations must be considered when interpreting these findings. The study is exploratory and based on a small sample, reflecting the practical challenges of recruiting non-speaking AAC users. Participants differed substantially in communication practices and levels of mediation, and findings are therefore interpreted as situated accounts rather than generalizable results. In addition, caregiver involvement, while often necessary, may have influenced how preferences were expressed, and the absence of video recordings limited analysis of non-verbal interactional nuances.

These limitations point to concrete directions for future research. First, the roles of caregivers and facilitators should be explicitly discussed and agreed upon prior to studies to balance accessibility with participant autonomy. Second, systematically documenting whether utterances originate directly from participants or are mediated could support more nuanced analyses of agency and authorship. Third, longitudinal studies that allow users to explore, test, and revise personalized voices over time and across everyday contexts are needed to better understand how preferences, trust, and voice ownership evolve.

7 Conclusions

This work contributes an exploratory qualitative account of how augmented communicators experience voice personalization as an interactive process. Using VoiceX as a design probe, we show that personalized voices matter not only for intelligibility, but for how users negotiate identity, agency, exploration, and control in AAC, revealing tensions between desire and resignation as well as moments of control and dependency shaped by caregiver mediation and AI systems.

Importantly, this poster does not evaluate VoiceX as a deployable AAC system, nor does it claim that AI-driven personalization inherently increases user agency. Instead, our findings highlight that agency in voice personalization is relational and negotiated, emerging through interaction between users, mediators, and technical constraints. Voice customization was experienced not solely as identity reproduction, but also as exploration, experimentation, and sometimes frustration.

By situating voice personalization within broader discussions of expressivity and authorship in AAC, this work reframes synthetic voices as one possible pathway rather than a universal solution for supporting identity expression. Future research should therefore move beyond one-off customization toward longitudinal, in-context studies that examine how personalized voices are adopted, revised, or rejected over time.

Rather than evaluating a system, this work highlights how voice personalization functions as a site of negotiation between identity, agency, and technical affordances in AAC.

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